DataCopilot – Project Charter

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Project Overview

DataCopilot will deliver a cloud-deployed web application, within eight weeks, that answers questions about CSV/Excel files and SQL databases in English or Hinglish with verified, explainable results.

Item	Details
Project title	DataCopilot – A Reliable Natural-Language Data Analysis System for CSV Files and SQL Databases
Program	Saras AI Institute – Capstone Project
Project lead	Mritunjay Sharma
Supervisor	Akshay Taneja, Professor of Practice (AI/ML)
Duration	8 weeks (Modules 1–8)

Goal: make data analysis accessible to non-technical users while reducing incorrect answers through SQL validation and answerability checking.

Approach: an LLM translates questions into SQL; a validator checks each query for schema correctness and read-only safety; DuckDB or the connected database computes the result; the system returns a chart, a plain-language answer, and the SQL used. Reliability will be measured on the BIRD benchmark and custom question sets.

Stakeholder Identification

Stakeholder	Role in project	Interest / expectation	Influence
Mritunjay Sharma	Project lead and developer	Deliver a working, well-documented system and research results	High
Akshay Taneja	Capstone supervisor	Feasible scope, quality deliverables, on-time submissions	High

Stakeholder	Role in project	Interest / expectation	Influence
Saras AI Institute	Program and evaluator	Project meets capstone learning outcomes	High
End users (small businesses, students, analysts)	Target users	Easy, trustworthy answers from their data	Medium
UAT testers (5+ volunteers)	Test the application in Module 7	Clear interface, accurate answers	Low
Open-source / research community	Future readers of the repository and paper	Reproducible code and evaluation	Low

Roles and Responsibilities

This is an individual project, so the project lead holds every delivery role; the supervisor and testers provide review and feedback.

Role	Person	Responsibilities
Project manager	Mritunjay Sharma	Planning, timeline tracking, weekly submissions, risk management
Data scientist / ML engineer	Mritunjay Sharma	Dataset profiling, benchmark creation, classifier training, evaluation and statistical testing
AI / LLM engineer	Mritunjay Sharma	Prompt design, SQL generation, validator, LLM comparison
Software developer	Mritunjay Sharma	FastAPI backend, Streamlit UI, charts, integration
Cloud / DevOps engineer	Mritunjay Sharma	Docker, AWS/GCP deployment, authentication, logging, monitoring
Technical writer	Mritunjay Sharma	Reports, user guide, GitHub documentation, research paper
Supervisor	Akshay Taneja	Approves topic and design, reviews deliverables, provides guidance
UAT testers	Volunteers (5+)	Use the deployed app and give structured feedback in Module 7

Key Milestones

Module	Milestone	Completion criteria
1	Proposal submitted	Topic approved; proposal, charter, architecture diagram and GitHub repository submitted
2	Solution design approved	Requirements, research review, detailed architecture and tech stack submitted; basic prototype runs CSV → SQL → answer
3	Data and benchmark ready	BIRD dev set loaded; CSV, Hinglish and unanswerable question sets created; EDA documented
4	Models evaluated	Answerability classifier trained and compared; baseline vs validated pipeline results recorded
5	Cloud deployment live	App running in Docker on AWS/GCP with login and logging
6	System integrated	End-to-end flow working: upload or connect → question → validated SQL → chart and answer; auto insights live
7	Testing complete	Functional, performance, security and UAT results documented; user guide and draft report ready
8	Capstone complete	Final report, presentation, live demo and complete GitHub repository submitted

Risks and Mitigation Strategies

The highest risks are model quality and schedule; each has a fallback that keeps the core project deliverable.

Risk	Likelihood	Impact	Mitigation
LLM generates incorrect SQL	High	High	Schema validation, read-only checks, and benchmark testing; this is the project's core research focus

Risk	Likelihood	Impact	Mitigation
Local 7–8B models too weak on BIRD	Medium	Medium	Compare several open models; use a free hosted API if needed; report relative improvement, not absolute score
Free-tier cloud server cannot host an LLM	High	Medium	Keep the app lightweight on the server; call an LLM API for the deployed demo
Free API rate limits during demo or UAT	Medium	Medium	Cache repeated answers; prepare a local fallback model
Schedule slips due to scope	Medium	High	Lock scope after Module 2; drop optional features in this order: full BIRD → BIRD subset → PostgreSQL → Hinglish
BIRD databases too large for laptop storage	Low	Medium	Use the dev set only; download early in Module 3
User data privacy (uploaded files)	Medium	High	Login required, read-only execution, temporary storage, no sharing of uploaded data
Unexpected cloud charges	Low	Medium	Budget alert set on day one; use only free-tier services; shut down after the capstone
Hinglish questions reduce accuracy	Medium	Low	Measure and report the gap; treat it as a research finding

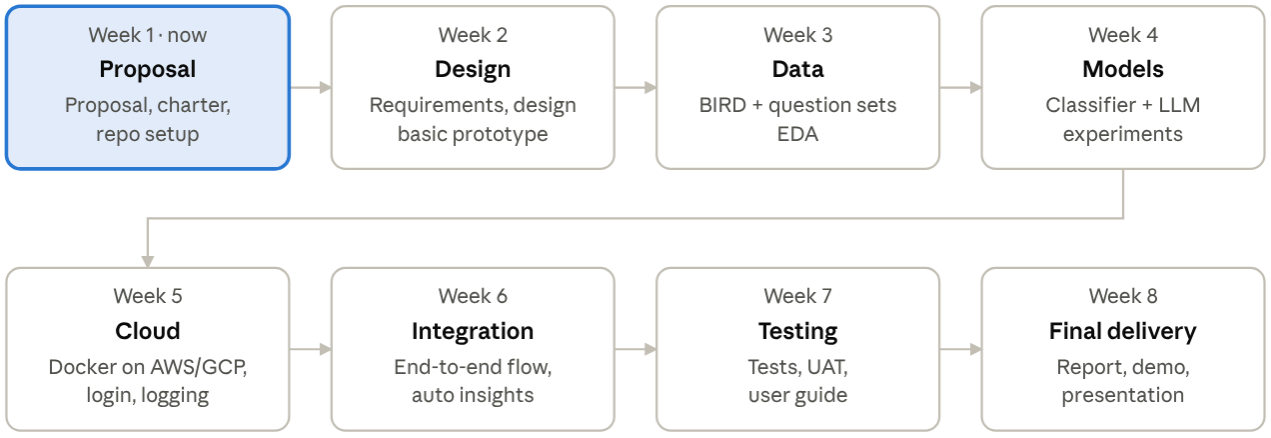
Communication Plan

What	With whom	Channel	Frequency
Module deliverables	Supervisor / evaluators	Saras AI portal	Weekly, per module deadline
Questions, approvals, blockers	Supervisor	Email	As needed, within 24 hours of a blocker
Progress evidence	Supervisor / evaluators	GitHub commits and README updates	Several commits per week

What	With whom	Channel	Frequency
Project updates and demos	Public / professional network	LinkedIn posts and short videos	Weekly (optional)
User testing feedback	UAT testers	Feedback form and short call	Once, during Module 7
Final presentation and demo	Faculty, peers, stakeholders	Live session	Module 8

Project Timeline

Eight one-week phases take DataCopilot from proposal to final demo



Project timeline · 8 weekly phases

Each phase matches one capstone module and ends with that module's portal submission. Coding starts in Week 1 so a working prototype exists before model development in Week 4.